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Deep Learning Breast Cancer Detection

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Introduction

According to the World Health Organization (WHO), breast cancer remains the most prevalent cancer among women in 157 countries. Early detection is key for improving survival rates and treatment outcomes; therefore, it is crucial to keep working on advancing diagnostic techniques.

Our project includes:

- **Classification** through a Convolutional Neural Network (CNN), specifically the ResNet architecture, to determine the presence of cancer and its type.
- **Segmentation** model using the U-Net architecture, to identify cancerous tissues within scans.

The dataset (downloaded from Kaggle) includes 780 grayscale images of breast cancer, captured over the course of about a year, obtained from Baheya hospital.

The images in the benign and malignant groups come with freehand segmentation masks made by radiologists at Baheya hospital.

Case	Number of images
Benign	487
Malignant	210
Normal	133

Classification

Data Transformation

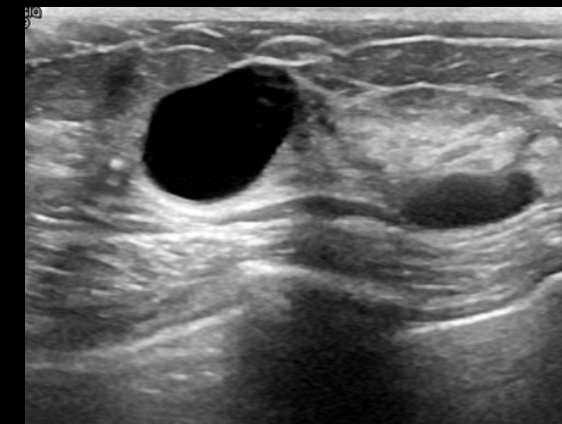
We implemented specific image transformations for both training and testing datasets to ensure consistency and increase model generalizability.

Training:

- **Resizing and Center Cropping:** all images are resized to 256 pixels and then center-cropped to 224 pixels, to ensure consistency in image dimensions across all inputs.
- **Data Augmentation Techniques:** random horizontal and vertical flips, random rotation. They reduce overfitting by increasing variability during training and improving robustness to unseen data.

Testing:

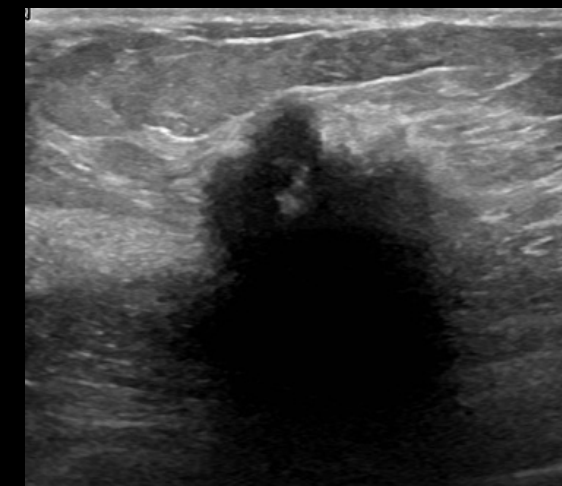
- **Resizing and Center Cropping:** all images are resized to 256 pixels and center-cropped to 224 pixels, in order to match the input size used during training.
- **No random augmentation:** to maintain consistency and control, no random augmentations are applied to the testing data.



Benign Image



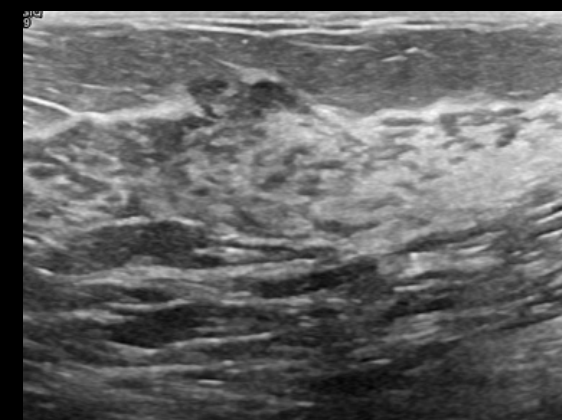
Mask Benign Image



Malignant Image



Mask Malignant Image



Normal Image

Normal Mask Image

Classification

ResNet18

Classification was performed using a pre-trained **Residual Neural Network (ResNet)** model.

One significant advantage of ResNet over traditional CNN models is its ability to mitigate the vanishing gradient problem (VGP), which can occur during training when gradients become too small to influence updates.

The VGP can adversely affect the learning capacity, slow down convergence, and hinder the network's ability to learn long-term dependencies.

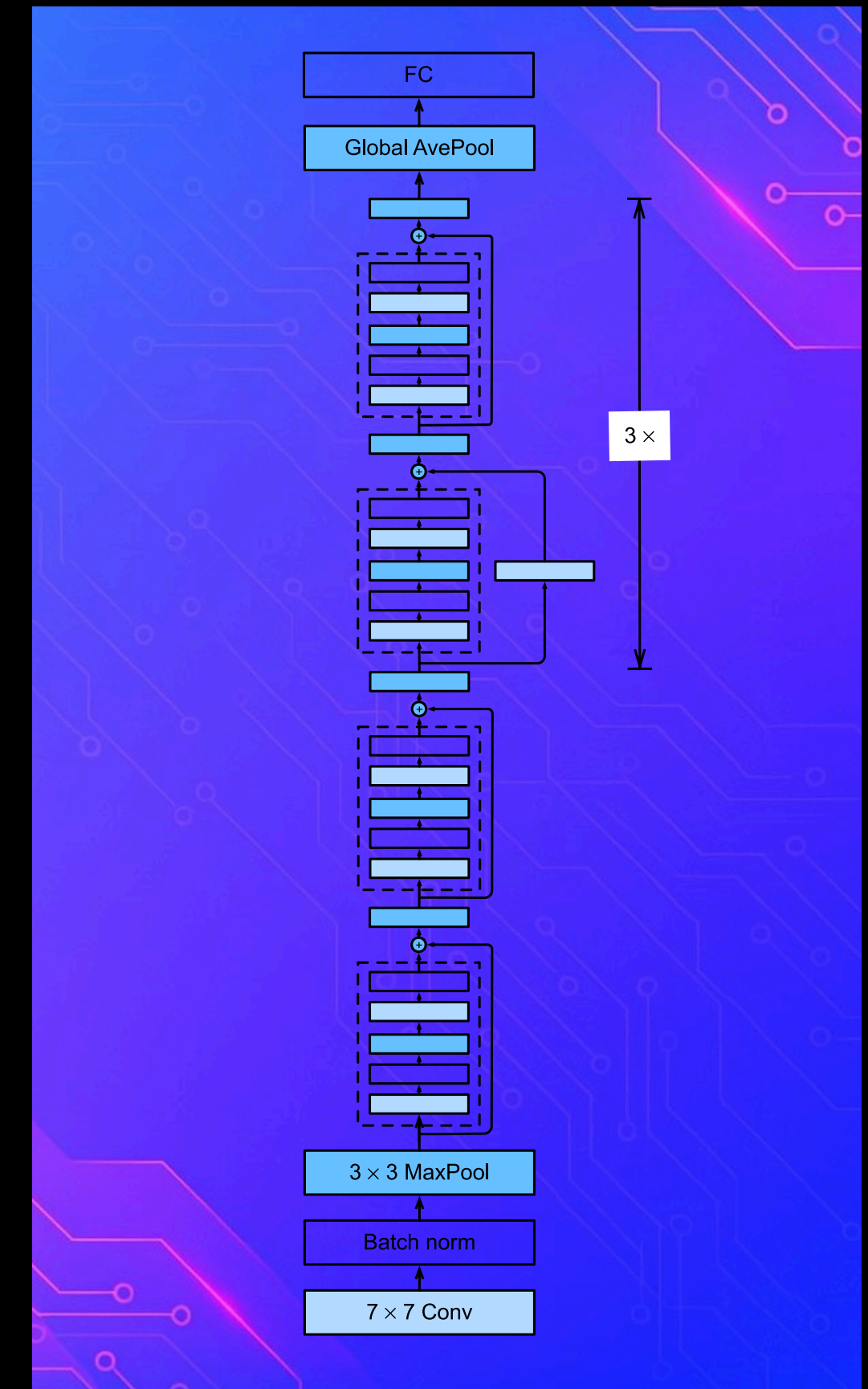
The key innovation of ResNet is the introduction of **Residual Connections**, which add a direct path for the gradient to flow through, bypassing one or more layers.

Specifically, in ResNet, a shortcut connection is added to each pair of convolutional layers, allowing the network to learn the residual mapping instead of the original mapping, facilitating easier optimization and improving performance.

We adapted the ResNet18 architecture to include a mask as an additional input channel, improving the network's ability to focus on relevant areas of medical scans.

To incorporate the mask, we implemented a custom dataset class (*ScanWithMaskDataset*) and modified the first convolutional layer of ResNet18 to accept four input channels (RGB + mask).

The final fully connected layer was adapted to predict three classes: *benign*, *malignant* and *normal*.



Classification

Model Training

Effective model training requires a careful selection of various components, including the Loss Function, Optimizer, and Learning Rate Scheduler.

Loss function

We implemented the **Cross Entropy Loss**, which is well-suited for multi-class classification tasks.

This function measures the discrepancy between the predicted class probabilities and the true labels, indicating how closely the model's predictions align with the actual distribution.

Optimizer

We selected the **Stochastic Gradient Descent (SGD)** with specific parameters:

- **Learning Rate** (0.001): controls how much the model's weights are adjusted in response to the error.
- **Momentum** (0.9): accelerates the optimizer, leading to faster convergence.
- **Nesterov Accelerated Gradient**: improvement compared to traditional momentum, computes the gradient at the anticipated position of the parameters.
- **Weight Decay** (0.00001): adds a regularization term to the loss function to prevent overfitting by penalising large weights.

Learning Rate Scheduler

This scheduler reduces the Learning Rate at specific intervals, which helps in fine-tuning the model as training progresses:

- **Step Size** (10): the interval (in epochs) at which the learning rate is adjusted.
- **Gamma** (0.1): the factor by which the learning rate is multiplied after each step size interval. Reducing the learning rate helps in achieving more precise convergence.

Classification

Model Training

The training process involves multiple iterations (epochs) where the model learns from the training data and its performance is evaluated on the validation data.

1. **Forward Pass:** for each batch of images and labels, the images are passed through the model to obtain predictions.
2. **Compute Loss**
3. **Backpropagation:** gradients are calculated by performing a backward pass. The gradients are then used to update the model's weights.
4. **Adjust Learning Rate**
5. **Validation:** every few epochs, the model's performance is evaluated on the validation dataset to monitor accuracy and adjust training strategies accordingly.

Classification

Model Evaluation - Evaluation Metrics

We proceeded by evaluating the model's performance on unseen data.

Predictions were made over the test set to compute key metrics, including loss, accuracy, precision, recall, and F1 score.

Test Loss	Accuracy	Precision	Recall (Sensitivity)	F1 Score
Measures the average Loss over the test dataset, indicating how well the model's predictions align with the true labels.	The proportion of correctly classified instances out of the total instances, providing a straightforward measure of performance.	The ratio of true positive predictions over the total predicted positives, indicating the model's ability to avoid false positives.	The ratio of true positive predictions over the total real positives, showing the model's ability to capture all relevant instances.	The harmonic mean of Precision and Recall, providing a balanced measure of the model's performance, especially in cases of class imbalance.

Classification

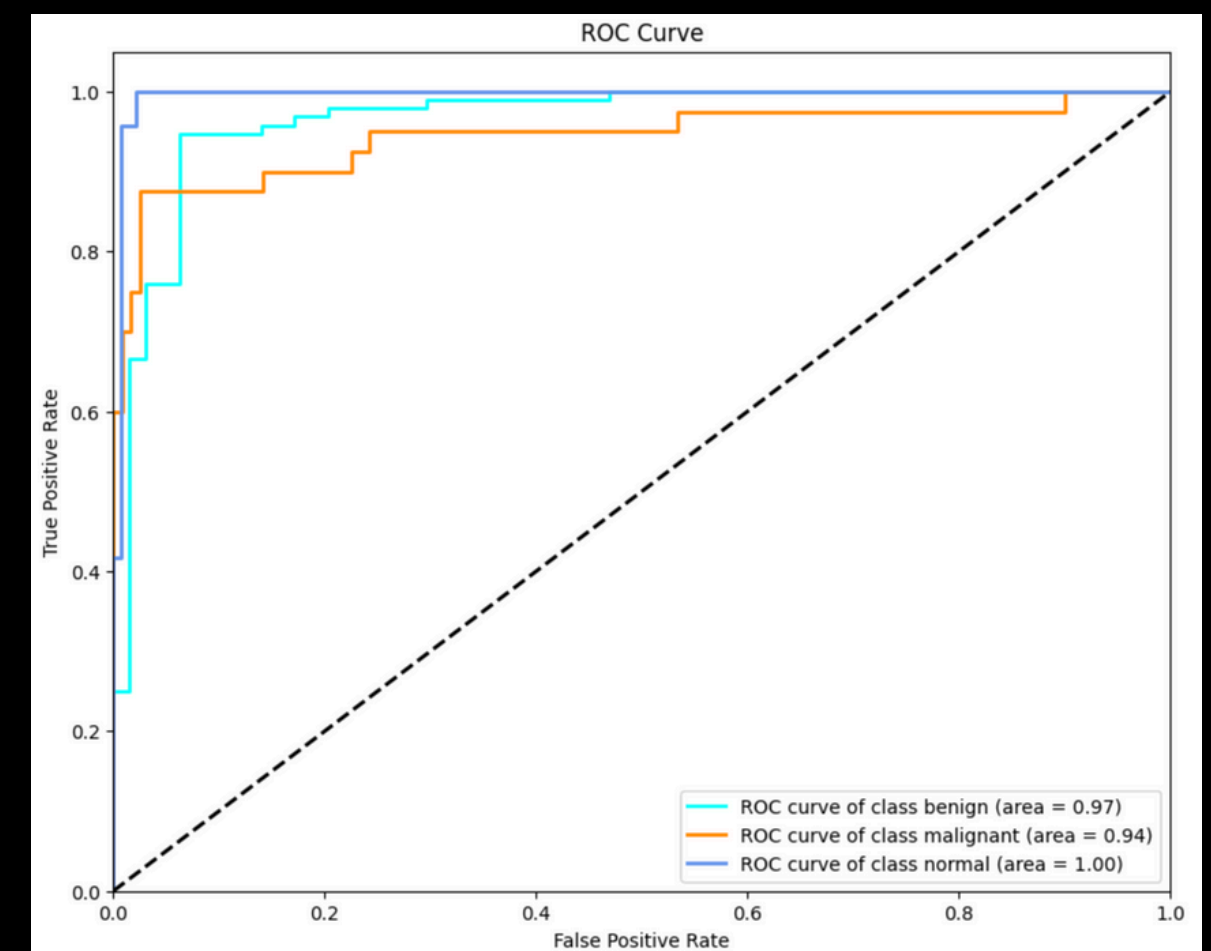
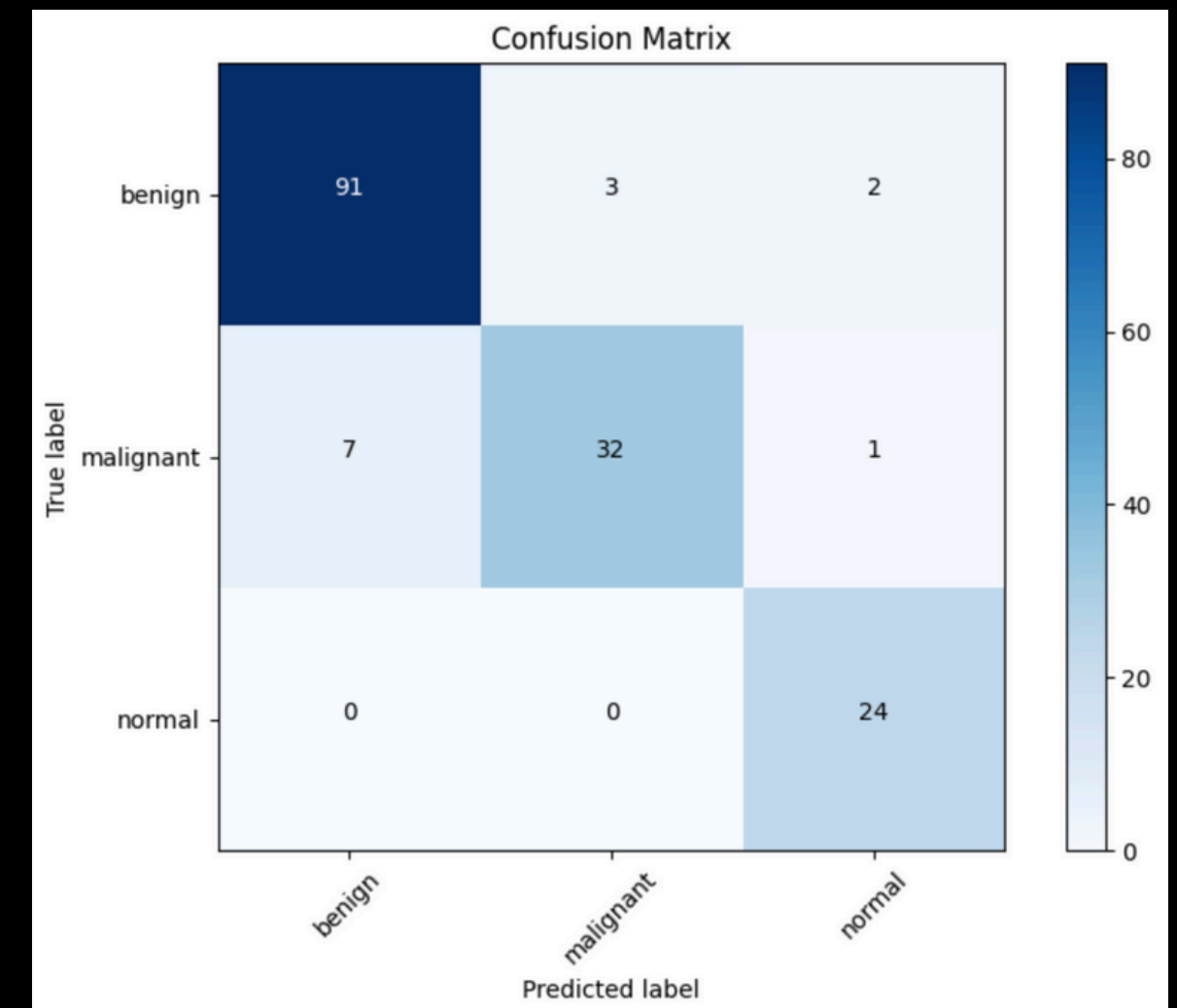
Model Evaluation

Confusion Matrix: summarizes the model's classification performance for benign, malignant, and normal cases by showing correct and incorrect predictions. It reveals that the model identifies normal scans with high accuracy and performs well on cancerous cases overall. However, some misclassification between benign and malignant classes remains, indicating a need for improved feature discrimination or further model tuning to better distinguish between these two conditions.

Receiver Operating Characteristic Curve (ROC): shows the performance of a classification model at all classification thresholds.

This curve plots two parameters: True Positive Rate and False Positive Rate. As the plot clearly shows, model performs very well. The area under the curve (AUC) for no cancer class is equal to one which suggests that our predictions are almost entirely correct. For benign and malignant classes, the AUC is equal to, respectively, 0.97 and 0.94.

In medical diagnostics false positives and false negatives have serious implications. FP can cause unnecessary stress and lead to additional tests, while FN are more critical because missing a cancer diagnosis can delay essential treatment. For this reason, achieving a balance between these two error types is essential, and the results of our study demonstrate a well-balanced performance.

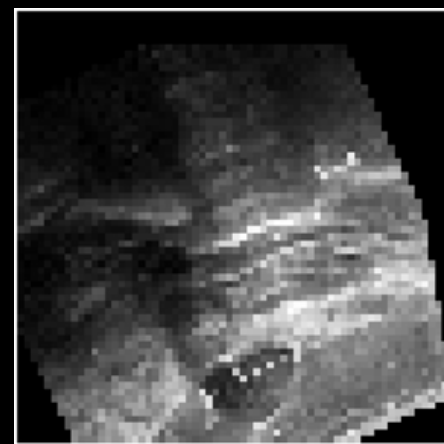


Segmentation

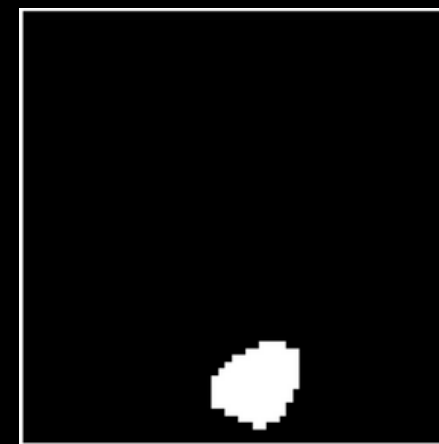
Introduction - Data Transformations and Dataset Creation

The goal is to identify and localize cancerous tissues at pixel level, providing spatial informations about the tumor. For the Segmentation task, only benign and malignant images were used, this reduced the dataset to 697 images, making generalization more challenging due to the limited data. In a real-world setting, using a pre-trained network would be a more effective approach; however, for the purposes of this project, we chose to train the model from scratch to evaluate its capabilities.

First of all we apply transformations to the images, using Data Augmentation Techniques, to ensure consistency and increase model generalizability. Then, we create a custom *SegmentationDataset* to ensure that each ultrasound image is correctly paired with its corresponding mask, this guarantees reliable data loading and prevents errors during training and evaluation. We also do a visual inspection of the samples of the segmentation dataset, in this way we verify that the image and mask are correctly aligned.



Image



Mask

Segmentation

U-Net

U-Net is a CNN architecture popular for image segmentation, and is perfect for medical images, because it works well with small datasets and locates even low-contrast structures.

It does not produce a class, but produces an output map, i.e., a **pixel-wise mask**. In fact, for each pixel of the image, U-Net decides: *lesion* or *background*.

There are two main components:

- The **Encoder** (contracting path): progressively extracts high-level features from the input image and reduces spatial resolution.
- The **Decoder** (expanding path): reconstructs the segmentation map by upsampling feature maps and progressively restores the original image resolution, to produce pixel-wise predictions.

A key feature of U-Net is the presence of **Skip Connections**, which directly connect encoder layers to their corresponding decoder layers, transferring high-resolution spatial information from the encoder to the decoder. They help to preserve object boundaries and improve localization accuracy.



Segmentation

Training Components

Loss Function: measures the difference between the predicted mask and the real mask, pixel per pixel.

We train a U-Net model for binary segmentation using **Binary Cross-Entropy with Logits Loss**, which combines a Sigmoid Activation and Binary Cross-Entropy.

For each pixel, we want to estimate how likely it is that it belongs to the lesion region, but neural networks do not directly produce the probability; in fact, the network's "raw" output is a logit, i.e., a real number. At this point, the Sigmoid transforms the logit into a value between 0 and 1, so that the output can be interpreted as a probability. Then, BCE measures how far the predicted probability is from the true value.

Optimizer: updates model weights to reduce Loss.

As in the Classification part, we select the **Stochastic Gradient Descent** (SGD) with Nesterov momentum (0.9), ensuring stable and smooth optimization. In addition, we add Weight Decay (1×10^{-5}) as regularization.

Learning Rate Scheduler: we select **Cosine Annealing**, where Learning Rate starts from an initial value of 0.1 and, following a cosine curve over 50 epochs, gradually decrease to 1×10^{-5} .

Segmentation

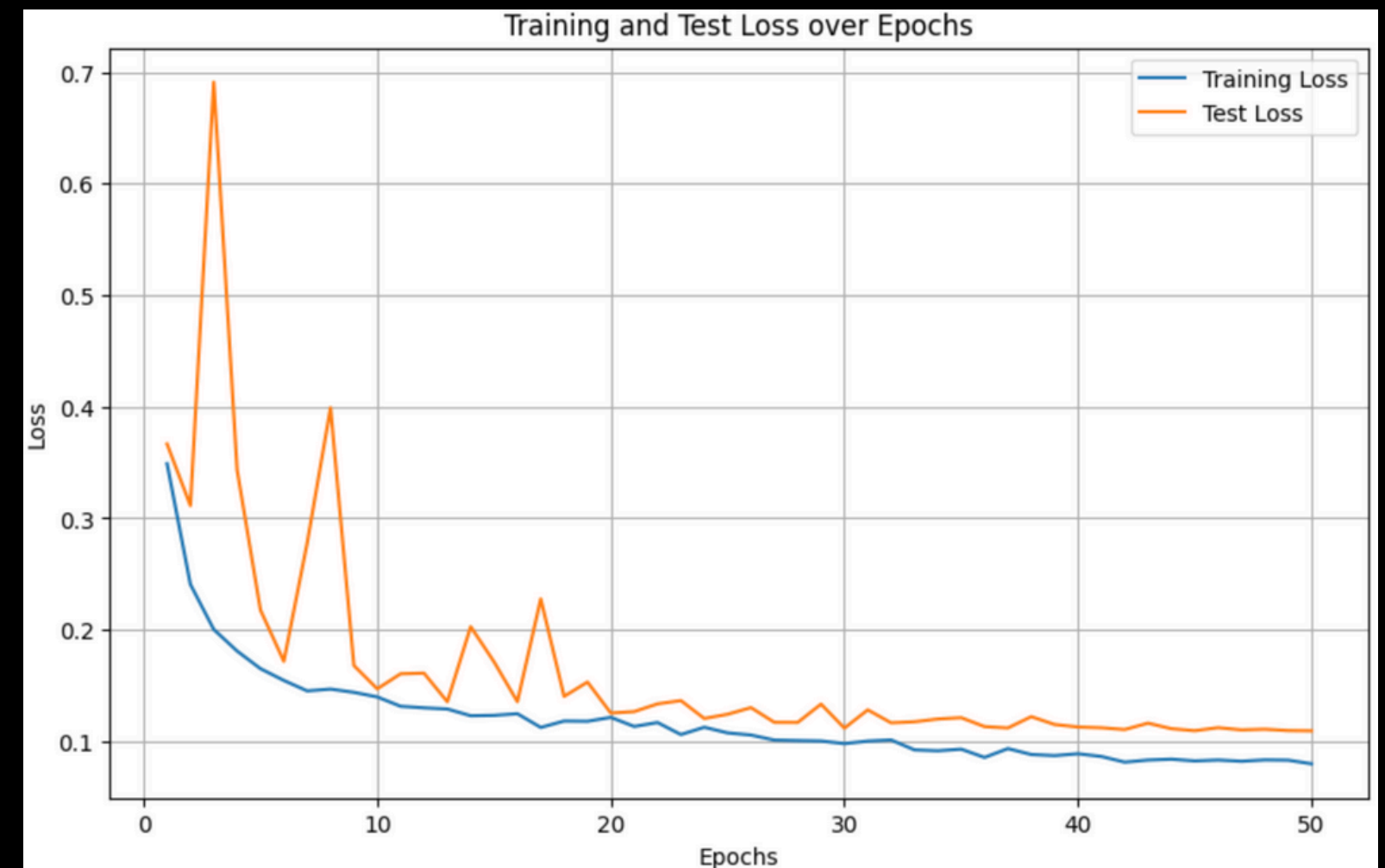
Training

During Training, both Training and Test Loss are monitored, to see how the model learns over time and how much it generalizes.

From the Loss curves, we observe that:

- An initial exploratory phase occurs at higher Learning Rates.
- After about 20 epochs, the training and validation Loss begin to decrease together.
- After 40 epochs, the model reaches a stable regime with low and constant errors.

This behavior indicates good convergence and no significant overfitting.



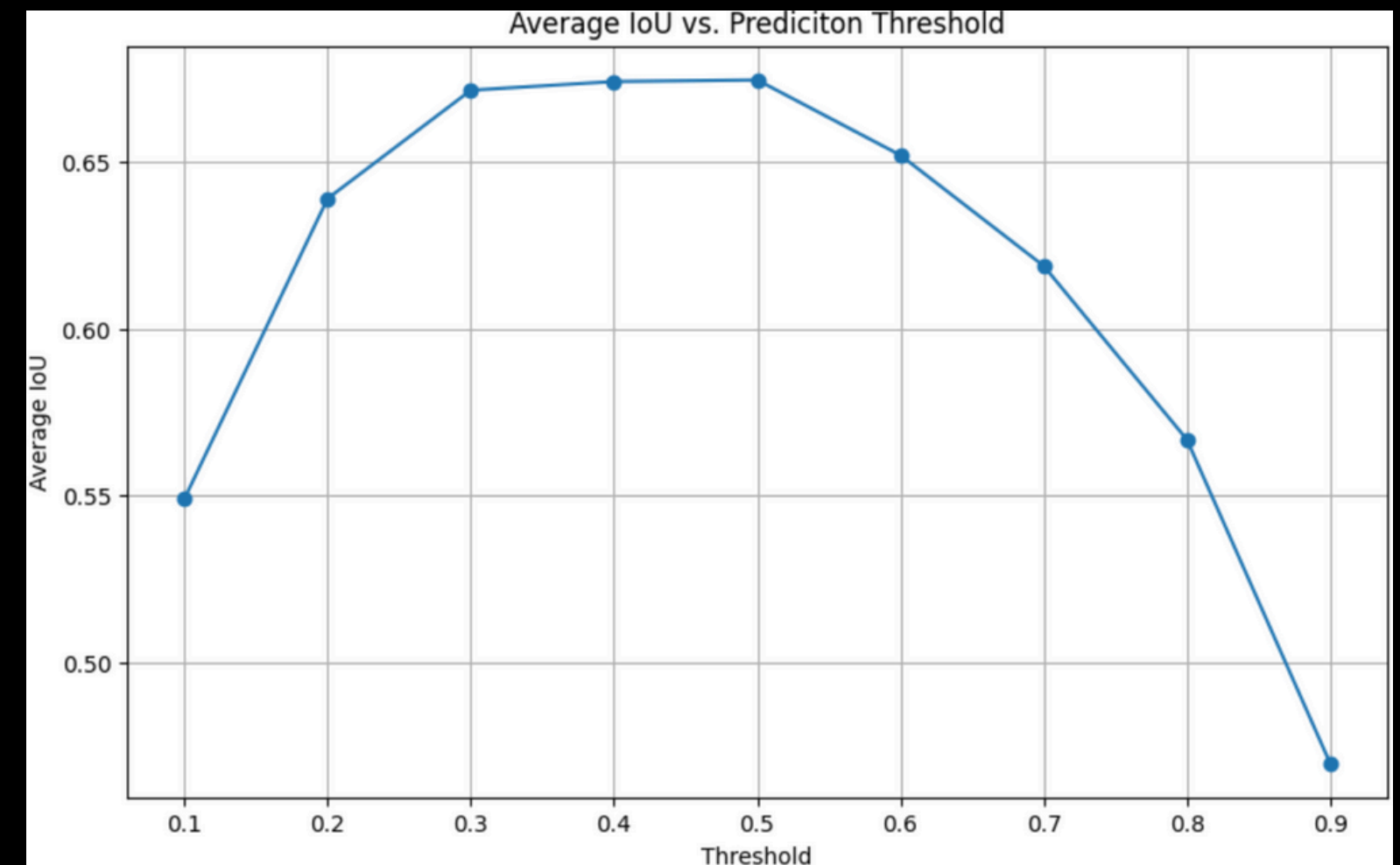
Segmentation

Inference

The network outputs a pixel-wise value in the range (0, 1), which can be interpreted as the probability that each pixel belongs to the *cancer region*. To perform Inference, a **threshold** is applied to these probabilities to convert them into a **binary segmentation mask**.

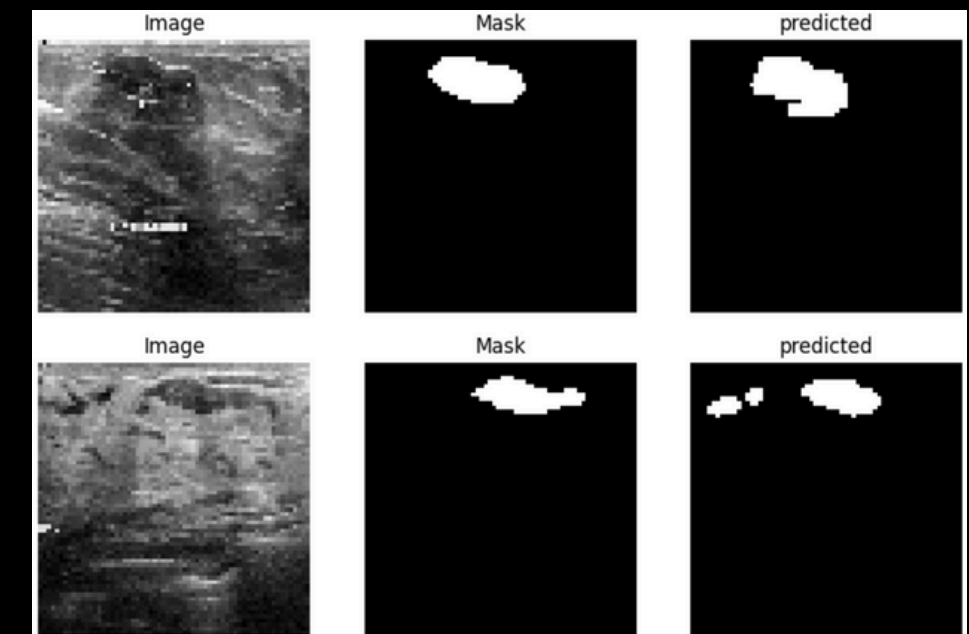
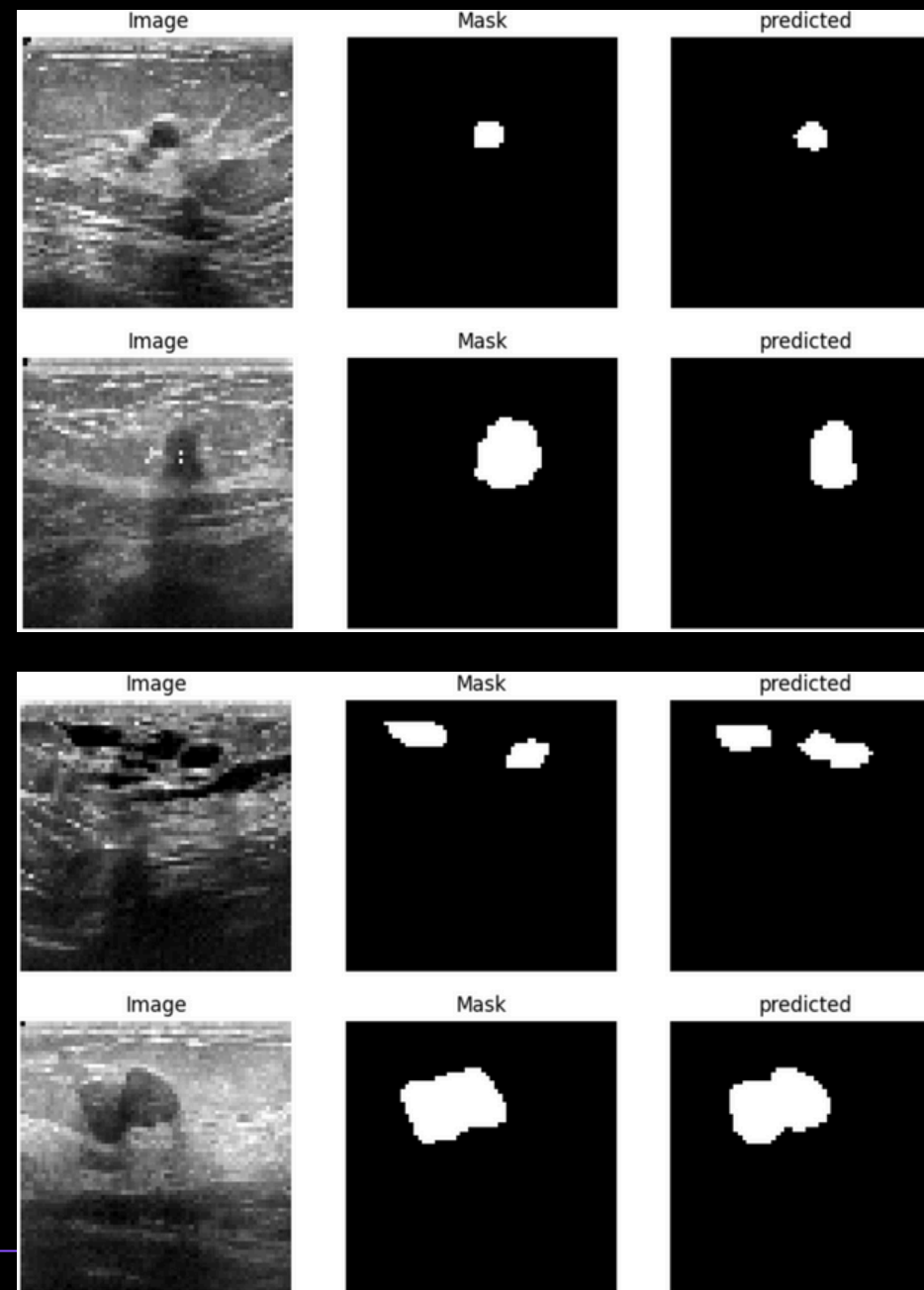
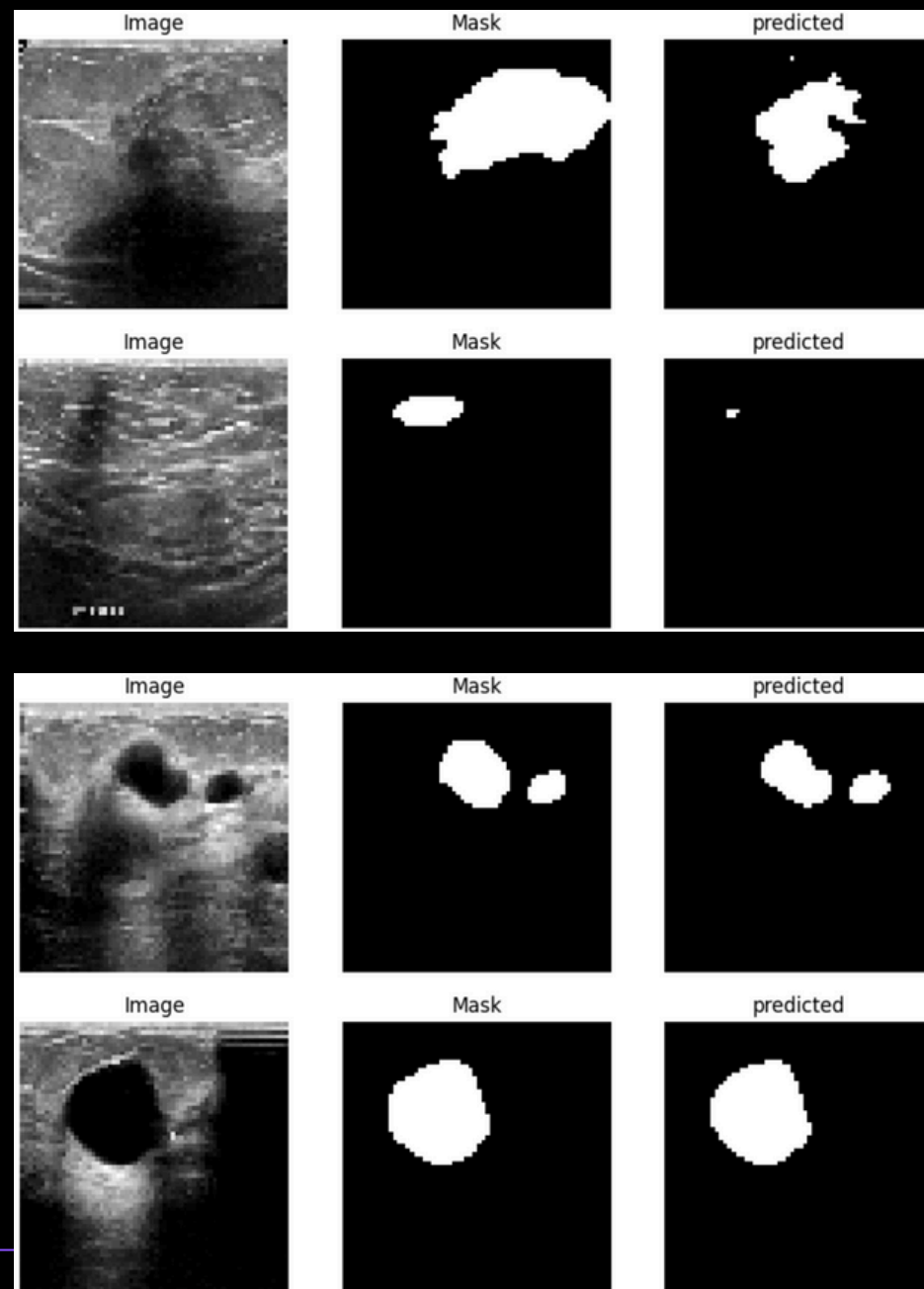
To select the optimal threshold, we evaluated the Segmentation performance using the average **Intersection over Union (IoU)**. IoU measures the overlap between the predicted mask and the ground truth, and is a standard metric for segmentation tasks.

By analyzing the average IoU for different threshold values, we observed that a threshold of 0.5 provides the best trade-off between false positives and false negatives, achieving the highest IoU.



Segmentation

Test Set



Overall, the predictions demonstrate that tumor regions are accurately localized and the predicted masks preserve the shape and extent of the lesions. The model generalizes well despite the relatively small dataset size.

Conclusion

Final Comments and Limitations

This project demonstrates the successful development and evaluation of a deep learning-based Classification and Segmentation model for breast ultrasound images.

The model achieves high Classification performance with strong generalization across classes. Despite minor confusion between benign and malignant cases, ROC analysis confirms reliable and well-balanced predictions, supporting the clinical relevance of the approach.

The qualitative results of the Segmentation model show that the U-Net model is able to accurately localize and segment breast lesions in most test images. Predicted masks closely match the ground-truth annotations in terms of shape and position, even for small or low-contrast lesions. Some minor discrepancies appear along lesion boundaries or in challenging cases with noisy backgrounds, which is expected in ultrasound imaging.

Overall, the results indicate good generalization and robust pixel-wise segmentation performance.

The limited size of our dataset restricts the model's ability to generalize. While data augmentation helped, it cannot fully replace the diversity of real-world data. Future research should focus on collecting larger datasets to improve robustness. Performance could also be enhanced by integrating multi-modal information, such as patient history or genetic data, alongside mammography images.